Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

Title: Creating Power BI Visualizations to Identify Key Business Patterns Across Products, Categories, and Regions

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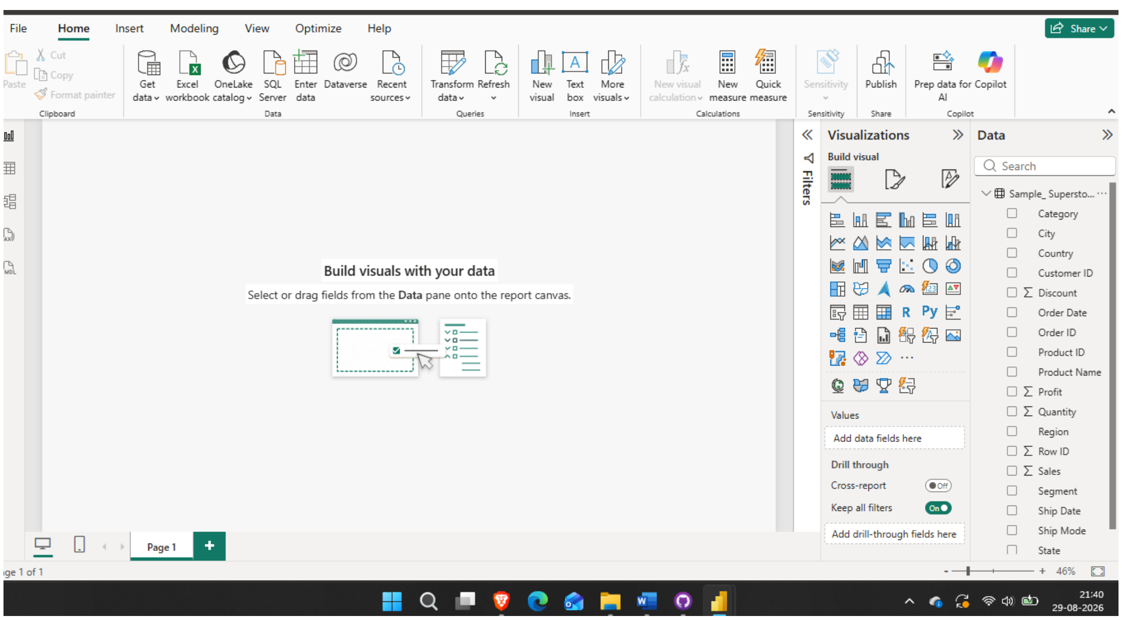
Creating Power BI Visualizations to Identify Key Business Patterns Across Products, Categories, and Regions

Problem Statement

The objective of this experiment is to design and build appropriate Power BI visualizations — KPI cards, trend charts, comparison charts, and interactive slicers — that help identify key business patterns such as top-performing regions and categories, loss-making products, and the relationship between discounting and profitability.

Step 1: Data Preparation

- Loaded Sample_Superstore.csv into Power BI Desktop using Get Data → Text/CSV.



- Identified inconsistent date formatting in the Order Date and Ship Date columns — some rows used a dash separator (e.g. 11-08-2016) while others used a slash separator (e.g. 6/16/2016), which was causing DataFormat.Error values during type conversion.
- Diagnosed the pattern by inspecting the raw values: both formats represented the same underlying month-day-year (US) convention and differed only in the separator character used, most likely introduced when the file was re-saved in Excel.

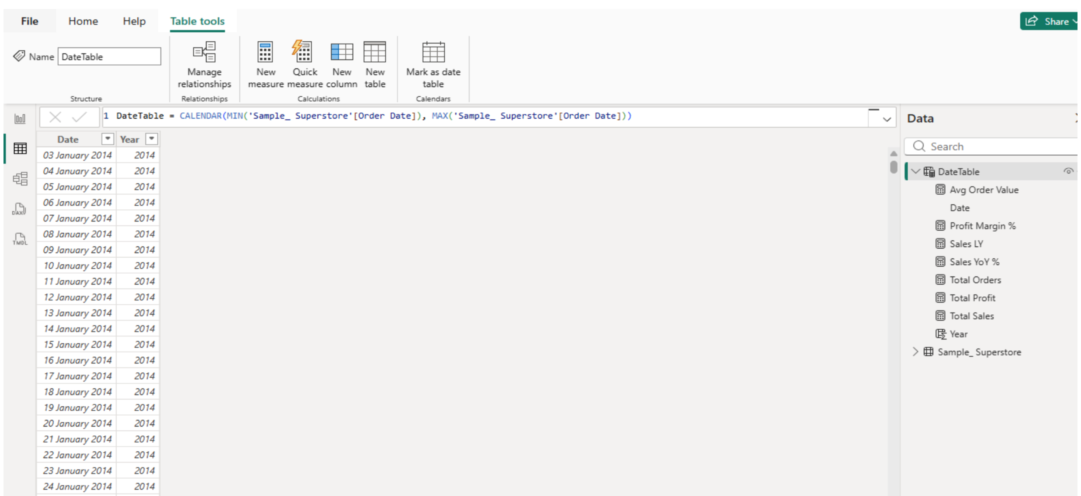
- Cleaned both date columns in Power Query using a custom column with the following M formula, which standardizes the separator before parsing:

```
Date.FromText(Text.Replace([Order Date], "-", "/"), "en-US")
```

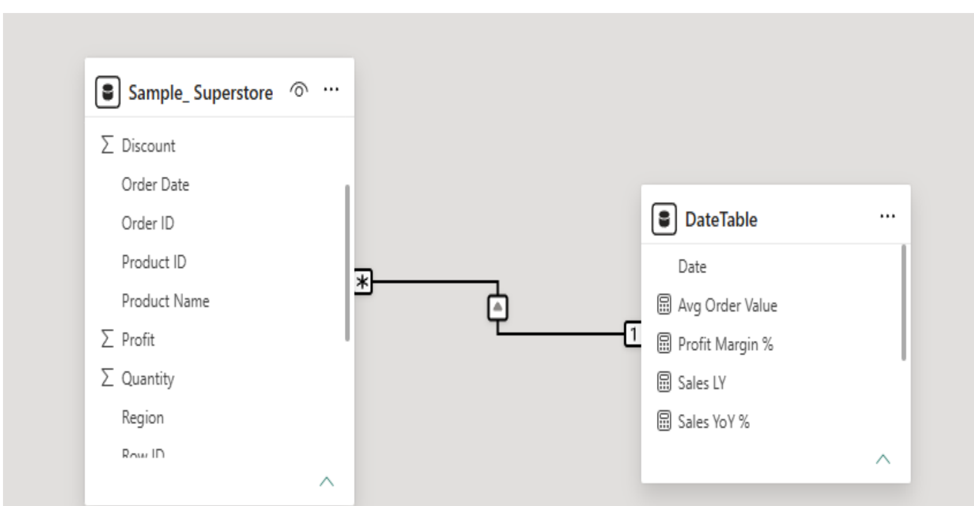
- Applied the same transformation to Ship Date. Column quality checks in Power Query confirmed 0% errors and 100% valid values for both date columns after the fix.

Step 2: Data Modeling

- Created a dedicated DateTable using the DAX formula CALENDAR(MIN(Sample_Superstore[Order Date]), MAX(Sample_Superstore[Order Date])), and marked it as the official Date Table in Table Tools.



- Added a calculated Year column to Date Table (Year = YEAR(Date Table[Date])) to support clean year-based slicing instead of a continuous date-range slider.

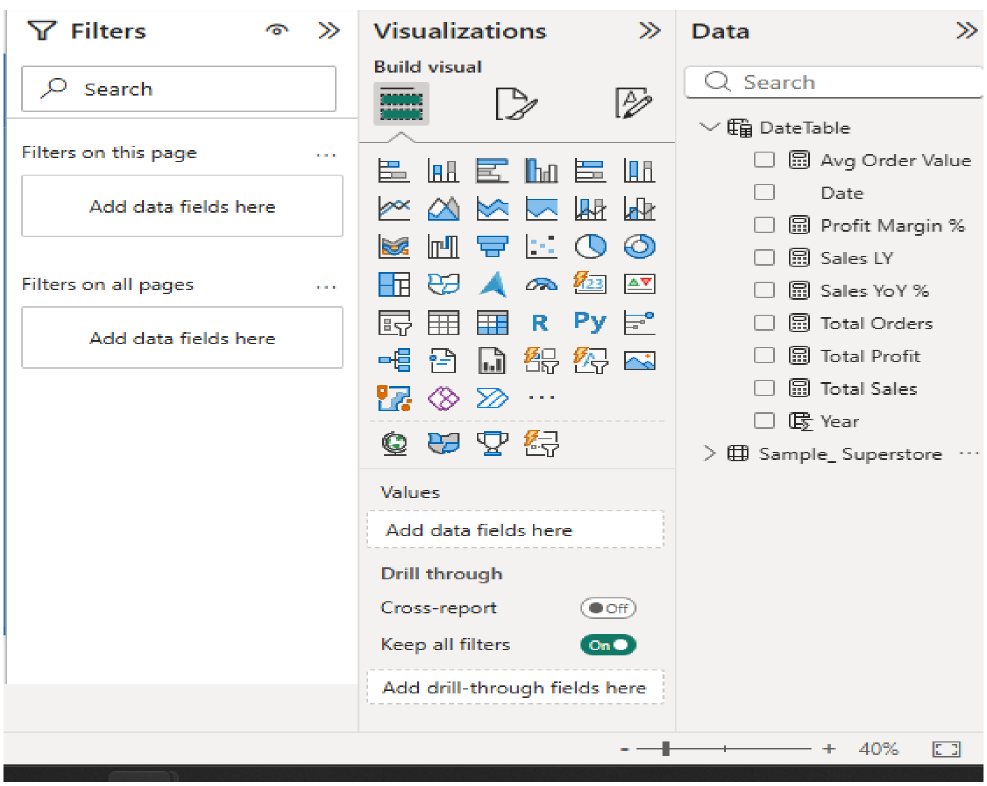


- Built a one-to-many relationship between Date Table[Date] (the "1" side) and Sample_ Superstore[Order Date] (the "*" side), and verified the cardinality and direction in Model view.

Step 3: DAX Measures

The following measures were created to drive the KPI cards and charts:

```
Total Sales = SUM(Sample_Superstore[Sales])
Total Profit = SUM(Sample_Superstore[Profit])
Profit Margin % = DIVIDE([Total Profit], [Total Sales], 0)
Total Orders = DISTINCTCOUNT(Sample_Superstore[Order ID])
Avg Order Value = DIVIDE([Total Sales], [Total Orders], 0)
Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR(DateTable[Date]))
Sales YoY % = DIVIDE([Total Sales] - [Sales LY], [Sales LY], 0)
```



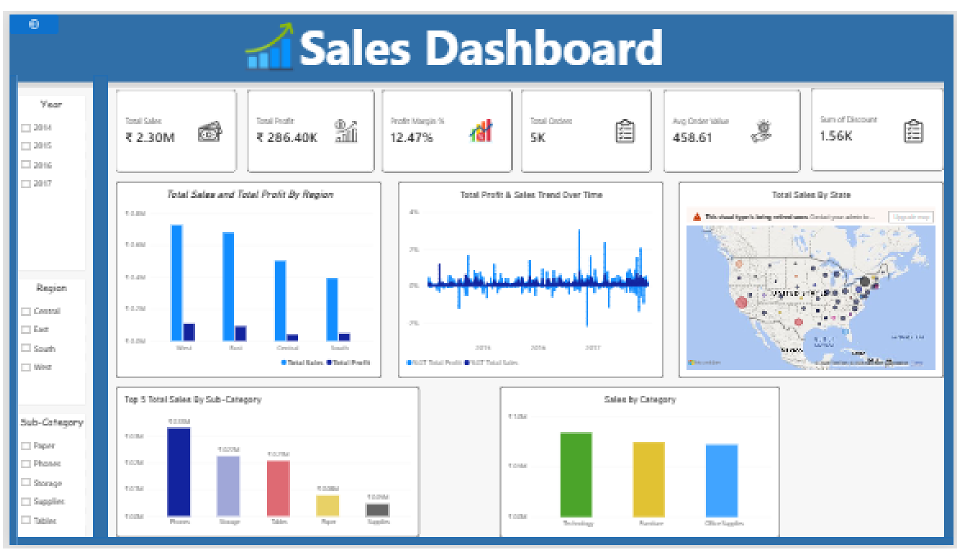
Step 4: Dashboard Development

- KPI Cards: Total Sales, Total Profit, Profit Margin %, Total Orders, avg Order Value, and Sales LY.
- Interactive Slicers: Year, Region, and Sub-Category, enabling management to filter every visual on the page simultaneously.
- Total Sales and Total Profit by Region — clustered column chart comparing revenue and profitability across Central, East, South, and West.
- Sales & Profit Trend Over Time — line chart built on Date Table, showing performance across 2014–2017.
- Top Sub-Categories by Sales — column chart ranking sub-categories by revenue contribution.
- Sales by Category — column chart comparing Furniture, Office Supplies, and Technology.
- Sales by Segment — donut chart comparing Consumer, Corporate, and Home Office customer segments.

Output

Data Model — Date Table related to Sample Superstore on Order Date (one-to-many):

Final Interactive Sales Dashboard



Key Business Patterns Identified

- West and East regions generate the highest total sales and profit; Central consistently lags behind the other three regions.
- Technology and its sub-categories (Phones, Copiers, Accessories) are the strongest contributors to both revenue and profit.
- Certain sub-categories, such as Tables and Bookcases, show weak or negative profitability despite reasonable sales volume, indicating a pricing or discounting issue.
- Profit tends to decline sharply as the average discount on a sub-category increases, with several sub-categories turning unprofitable beyond a certain discount level.
- Sales show a clear seasonal spike toward the fourth quarter (October–December) in each year of the dataset.
- The Consumer segment contributes the largest share of total sales among the three customer segments, followed by Corporate and Home Office.

Conclusion

This experiment used the Sample Superstore dataset to design an interactive Power BI dashboard capable of surfacing key business patterns across products, categories, and regions. Beyond visualization design, the experiment required diagnosing and resolving a real-world data quality issue — inconsistent date formatting — through Power Query and building a correct date-based data model to support time intelligence. The resulting dashboard combines KPI cards, trend analysis, and comparison charts with interactive slicers, giving management a fast, drillable view of overall business performance.

Learning Outcomes

Through this experiment, I learned:

- How to diagnose and resolve inconsistent date formats in a raw dataset using Power Query and custom M code.
- How to build a dedicated Date table and establish a correct one-to-many relationship to support time-based analysis.
- How to write DAX measures for KPIs, year-over-year comparisons, and profit margin analysis.
- How to design an interactive, management-ready dashboard using KPI cards, slicers, and multiple chart types (bar, line, scatter, donut).
- How to interpret visual patterns in sales data to draw concrete, actionable business insights.